Competitive Programming

1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

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n Lines - n Elements

Output Format:

Element x - That is repeated

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n,a[100];
    scanf("%d",&n);
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]==a[j]){
                printf("%d",a[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

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Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                printf("%d",arr[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Find the intersection of two sorted arrays.

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OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

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Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int t;
    scanf("%d",&t);
    while(t--){
    int n,m;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&m);
    int b[m];
    for(int i=0;i<m;i++){
        scanf("%d",&b[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            if(a[i]==b[j]){
                printf("%d ",a[i]);
                break;
            }
        }
    }
    printf("\n");
}
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

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The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int t;
    scanf("%d",&t);
    while(t--){
        int n,m;
        scanf("%d",&n);
        int a[n];
        for(int i=0;i<n;i++){
            scanf("%d",&a[i]);
        }
        scanf("%d",&m);
        int b[m];
        for(int i=0;i<m;i++){
            scanf("%d",&b[i]);
        }
        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){
                if(a[i]==b[j]){
                    printf("%d ",a[i]);
                    break;
                }
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

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Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n,k;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&k);
    int i=0,j=1;
    while(i<n && j<n){
        if(a[j]-a[i]==k){
         printf("1");
         return 0;
        }
        if(a[j]-a[i]<k){
            j++;
        }
        else{
            i++;
        }
        if(i==j){
            j++;
        }
    }
    printf("0");
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

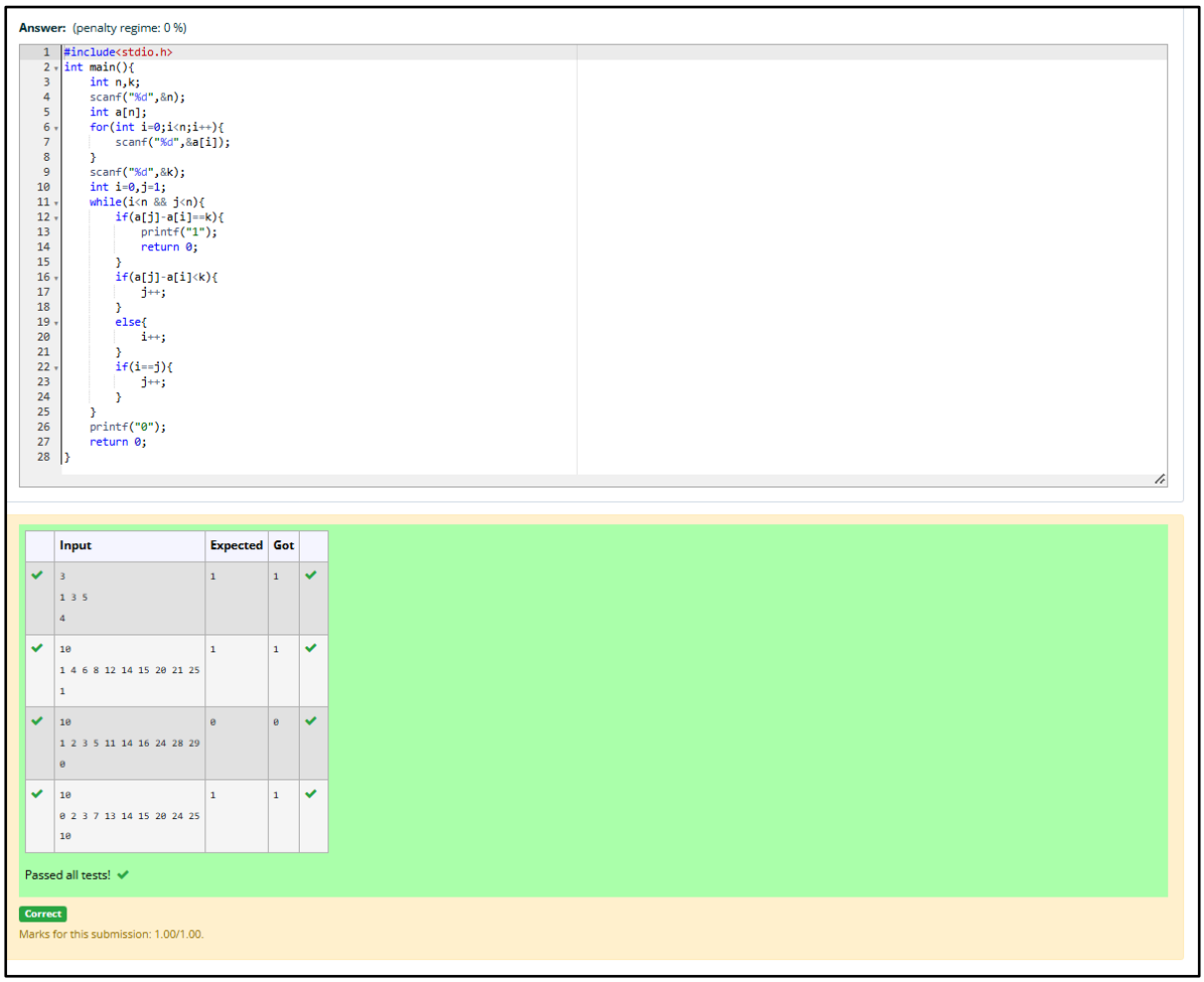
1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.



Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n,k;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&k);
    int i=0,j=1;
    while(i<n && j<n){
        if(a[j]-a[i]==k){
            printf("1");
            return 0;
        }
        if(a[j]-a[i]<k){
            j++;
        }
        else{
            i++;
        }
        if(i==j){
            j++;
        }
    }
    printf("0");
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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